

freedom subsystems can be solved independently. In this section we examine the stick-fixed free response associated with linearized, longitudinal rigid-body motion. These results provide a reasonable description of longitudinal aircraft dynamics provided that the aircraft is not undergoing large-amplitude oscillations or rapid maneuvers.

The longitudinal motion comprises two translational degrees of freedom (axial and normal) and one rotational degree of freedom (pitch). Since this motion has three degrees of freedom, it is characterized by six eigenvalues. The eigenvalues associated with the linearized longitudinal motion of a typical airplane contain two complex pairs, corresponding to two oscillatory modes. One of these modes is lightly damped with a period on the order of 30 seconds or more and is called the *long-period* or *phugoid mode*. The other oscillatory mode is heavily damped with a much shorter period and is simply called the *short-period mode*. The remaining two longitudinal eigenvalues are always zero, corresponding to simple rigid-body displacement modes.

The free response of the longitudinal system is obtained from the homogeneous equations, with the control deviations set to zero. As indicated in Eq. (7.7.9), for all practical purposes, four of the derivatives with respect to the translational accelerations are zero,

$$R_{x,\hat{\mu}} \cong 0$$

$$R_{z,\hat{\mu}} \cong 0$$

$$R_{m,\hat{\mu}} \cong 0$$

$$R_{x,\hat{\alpha}} \cong 0$$

From Eq. (7.7.5), the homogeneous linearized longitudinal equations can then be written in the form

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & (1 - R_{z,\hat{\alpha}}) & 0 & 0 & 0 & 0 \\ 0 & -R_{m,\hat{\alpha}} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta\hat{\mu} \\ \Delta\hat{\alpha} \\ \Delta\hat{q} \\ \Delta\hat{\xi}_x \\ \Delta\hat{\xi}_z \\ \Delta\hat{\theta} \end{bmatrix} - \begin{bmatrix} R_{x,\mu} & R_{x,\alpha} & R_{x,\bar{q}} & 0 & 0 & -R_{gx} \cos \theta_o \\ R_{z,\mu} & R_{z,\alpha} & (1 + R_{z,\bar{q}}) & 0 & 0 & -R_{gx} \sin \theta_o \\ R_{m,\mu} & R_{m,\alpha} & R_{m,\bar{q}} & 0 & 0 & 0 \\ \cos \theta_o & \sin \theta_o & 0 & 0 & 0 & -\sin \theta_o \\ -\sin \theta_o & \cos \theta_o & 0 & 0 & 0 & -\cos \theta_o \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta\mu \\ \Delta\alpha \\ \Delta\bar{q} \\ \Delta\xi_x \\ \Delta\xi_z \\ \Delta\theta \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (8.2.1)$$